WASP-10b

R-0820

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-10_171030` · created 2026-07-28T14:26:51+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458056.7609 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.156 +/- 0.011
Transit depth (Rp/Rs)^2	2.45 +/- 0.35 [%]
Orbital Inclination (inc)	89.82 +/- 0.58
Airmass coefficient 1 (a1)	0.986 +/- 0.013
Airmass coefficient 2 (a2)	0.011 +/- 0.01
Scatter in the residuals of the lightcurve fit is	1.27 %
Best Comparison Star	#1 - [268, 406]
Optimal Aperture	3.16
Optimal Annulus	11.76
Transit Duration (day)	0.0945 +/- 0.0012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.156 ± 0.011 vs 0.1592 (deviation 0.29σ)
Duration fit vs published	0.0945 d vs 0.0946 d (deviation 0.07σ)
Mid-time O–C	0.2 min
Depth SNR	14.2
Residual scatter	1.27 %

Light curve

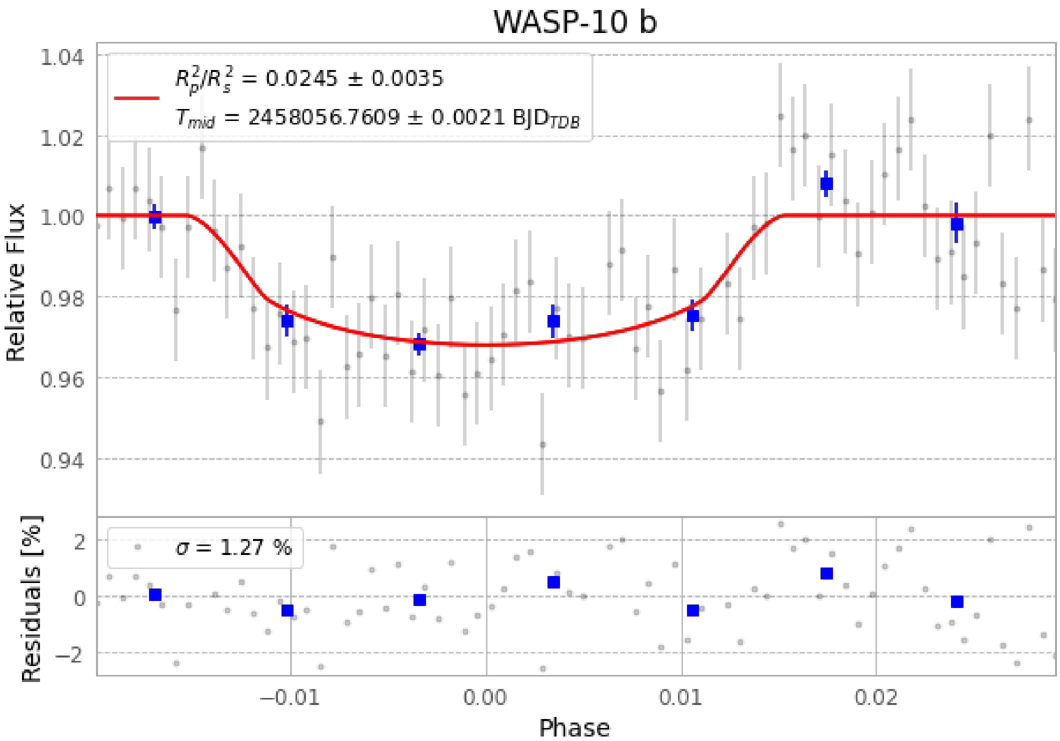

Figure 1 — FinalLightCurve_WASP-10 b_2017-10-29.png (SHA-256 561b9cca7f19fefe…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [274, 279], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 b5f031b0e0d77945…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

74 FITS frames (49 MB), `WASP-10171030043912.FITS` → `WASP-10171030081812.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-171030.log (bdbf0a07ca48…), FinalLightCurve_WASP-10 b_2017-10-29.png (561b9cca7f19…), FinalLightCurve_WASP-10 b_2017-10-29.csv (dc26dc8016b2…)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-28 14:26Z.